

Python Code

```
1  import nltk
2  from nltk import CFG
3  from nltk.parse import ChartParser
4
5  # Define the grammar
6  grammar = CFG.fromstring("""
7  S -> NP VP
8  VP -> V NP | V NP PP
9  NP -> Pronoun | Det N | Det N PP
10 PP -> P NP
11
12 Pronoun -> 'I'
13 V -> 'shot'
14 Det -> 'an' | 'my'
15 N -> 'elephant' | 'pyjamas'
16 P -> 'in'
17 """)
18
19 # Input sentence
20 sentence = "I shot an elephant in my pyjamas".split()
21
22 print("Sentence:", " ".join(sentence))
23
24 # Create parser
25 parser = ChartParser(grammar)
26
27 # Generate parse trees
28 trees = list(parser.parse(sentence))
29
30 print("Number of parse trees found:", len(trees))
31
32 # Display parse trees
33 for i, tree in enumerate(trees, 1):
34     print(f"\nParse Tree {i}:")
35     print(tree)
36
37 # Optional: Draw parse trees graphically
38 for tree in trees:
39     tree.draw()
```